

Split-zero structures and the zeta-function programme

A mathematical exposition with human-source provenance

An AI-assisted companion to the KokunoYumeto / The Clankers programme

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Fixed mathematical baseline: the accepted 726-page edition

Abstract

This companion explains a research programme connecting support-sensitive scalar structures, a classical theta quotient, complete zero jets, exponential periods, and determinant estimates. It defines the maps that join these constructions and identifies their human mathematical sources. The programme's theta quotient is explicitly identified with Ralf Meyer's quotient; its relationship to the geometric results of Pierre Deligne is stated with the relevant hypotheses. The current determinant calculation and its remaining target are written on their original spaces. The companion retains the complete proof archive and adds an entry route, source attributions, and a record of what has and has not been checked. The programme remains open; no proof of the Riemann hypothesis is claimed.

Purpose and reading conventions

Human readability and accurate provenance are commitments of this programme. Its mathematics should be pedagogically articulated so that people can understand and examine it. Its source record should fairly credit the humans whose work it uses. The Leiden Declaration and the September Math and AI declaration provide context for these editorial concerns (Alper et al., 2026; Avila et al., 2026). Neither declaration supplies mathematical hypotheses or endorses this programme.

The intended reader knows linear algebra, basic complex analysis, and the language of vector spaces and quotients. The analytic and categorical parts introduce their objects before using them. The appendix spells out the support, complex, and source-lifting maps that connect the opening definitions to the later finite calculations. A reader chiefly interested in the current estimate can follow the main text's period and metric constructions, then return to that appendix for the analytic realization.

For stable historical page references, **R14** refers to the complete editable source `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex`; **R09** refers to `09_UPDATED_JOINT_NOTE.tex` in the same accepted edition. Page numbers attached to R14 refer to the printed pages of the preserved 692-page predecessor reader, *Original Mixed Cohomology: Periods, Kernels and Arithmetic Control*. The version is available with complete public sources in the [17 September growing-degree release](#). Its source commit is `de0644902496e1718879f4b625f9ff9faafc5ae5`. These fixed predecessor locators remain meaningful. The accepted successor, **R726**, contains 726 pages and preserves every predecessor source byte while inserting the new proofs and their earlier receiving calculations. Its source commit is `be79088cb1fa4d79b31724856eefcddb0839b8cc`; its publication is [DOI 10.5281/zenodo.22803503](https://doi.org/10.5281/zenodo.22803503). Section 12 explains its advance with R726 locators.

Results imported from that archive are identified by their proof locations. Elementary comparison statements established here carry their proofs. References to the literature identify what is inherited, with conventions and hypotheses wherever those affect the map. A proved project calculation is not thereby a claim of historical novelty. Exact source-verification levels and unresolved bibliographic trails are retained in the accompanying human-source guide and attribution records.

The originating programme is directed under the public name KokunoYumeto. Its notes and this companion use AI assistance. The accompanying disclosure records that role separately from the authors of the mathematical literature. No independent human peer review of this companion is asserted. The reader should judge the definitions, proofs, source comparisons, and scoped evidence provided here and in the complete archive.

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1 The question and the present answer

The programme asks whether a cohomological construction over a proposed absolute base can retain the arithmetic information of the Riemann zeta function and provide sufficiently strong control of a specified spectral action. The current calculations study the consequences of a hypothetical off-critical-line zero quartet. They build the associated spaces and maps, preserve repeated zeros, and calculate norms and determinant returns on those actual spaces. This is research toward a geometric argument about zeta zeros. The edition proves a substantial collection of finite-dimensional identities and estimates. It does not prove the Riemann hypothesis, establish the existence of an off-line quartet, or exclude such a quartet.

The inspiration from Pierre Deligne is precise at the level of ambition: a cohomological realization should come with a structure that controls spectral size. Deligne's theorem for smooth projective varieties over finite fields has an actual variety, an actual Frobenius action and a purity statement. Here the base, support structure, analytic classes, periods and metric comparisons have to be constructed and connected. Their formal resemblance to finite-field geometry is a reason to calculate those connections, not a proof of a purity theorem. The original audit constructs specific scaling and nilpotent maps and records the differences between them. Deligne's *La conjecture de Weil. I*, theorem 1.6, supplies that finite-field statement [Deligne \(1974\)](#). His later theory of weights is a distinct development [Deligne \(1980\)](#). Neither statement is silently imported into the present analytic construction.

The central allocation problem, retained in the 726-page successor, is concrete. A known total determinant growth splits into two correlated contributions. Their sum has been evaluated. Their separate leading allocation has not. The surviving expression is a signed spectral profile of eight specified four-by-four positive matrices, together with four specified complementary quotient returns. The matrices are computed from the original period, Gamma recurrence, conductor division and lower-evaluation minimum; they are not arbitrary test matrices. A later part of this exposition writes down their exact formula.

There is a second, related calculation concerning the original action's canonical allowance. Strong lower bounds for that allowance have now been proved, including growing polynomial-degree windows. An improvement to the mixed allocation does not automatically improve that action bound: an exact cancellation must be applied first. Keeping these two questions connected through that identity is essential to understanding both what has been achieved and what remains.

2 What the support base is recording

The base notation distinguishes *absence* from a *present coefficient equal to zero*. This project construction is documented in [ChatGPT 5.4 and Gemini 2.5 DeepThink \(2026\)](#); [The Clankers \(2026\)](#); its human mathematical background and exact maps are developed in [Appendix A.1](#). For a ring R , the underlying carrier is

$$G(R) = \{\tau\} \sqcup R^\bullet, \quad e_R = 0_R^\bullet.$$

The two observations are

$$(p_R, b_R)(\tau) = (0, 0), \quad (p_R, b_R)(r^\bullet) = (r, 1).$$

Their joint map is injective. Indeed two present points with the same first coordinate have the same coefficient; an absent and a present point have different second coordinates. Thus

$p_R(e_R) = p_R(\tau) = 0$ does not identify e_R with τ . The support bit records a distinction that the scalar observation forgets. The proposed absolute base is denoted $\mathbf{F}_{1,\tau} = \{\tau, 1\}$, with its specified pointed/preaddition structure. A structural map sends its absent element to τ and its unit to $1_R^\bullet$. The structural base map belongs to the pointed/preaddition category; the coefficient adjunction belongs to the diagram category. Their complete definitions and proofs are retained in the original base-formation source.

Why does this distinction survive so far into an analytic calculation? Because a relation may map to zero *at an existing support label*. For a coefficient map L , its lift on a present label ω is

$$(\omega, x) \mapsto (\omega, Lx).$$

If $Lx = 0$, the image is $(\omega, 0)$, not external absence. When a larger coefficient face is obtained by zero insertion j_A^B , the actual compatibility required is $L^B j_A^B = j_A^B L^A$. This is a commutative-square statement about the maps. It preserves the original outer leg and every coefficient face; it does not promote a coefficient-linear section to a unital algebra morphism. [R09, §1, “Scope, original input, and retained support”; R14, AKP1–5, §26, pp.110–111. The full categorical definitions and proofs are in the original base-formation audit, chapter 01, §§1–4.]

3 The analytic source: theta, Mellin transform and full jets

Write the Riemann xi function in the convention

$$\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s), \quad g(s) = 2\xi(s).$$

The Riemann zeta and theta setting is classical [Riemann \(1859\)](#). The original test space and theta map are

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = 0, \int_{\mathbb{R}} \phi(x) dx = 0\}, \quad \Theta\phi(x) = 2 \sum_{n \geq 1} \phi(nx).$$

The target $\mathcal{B}$ consists of smooth functions on $\mathbb{R}_{>0}$ for which

$$\sup_{x>0} x^b |D^j F(x)| < \infty \quad (b \in \mathbb{Z}, j \geq 0), \quad D = -x\partial_x.$$

Its Mellin transform is $\mathcal{M}F(s) = \int_0^\infty F(x)x^s dx/x$. Integration by parts, whose two boundary terms vanish by the defining decay at zero and infinity, gives

$$\mathcal{M}(DF)(s) = s\mathcal{M}F(s).$$

This is why a polynomial in the Euler operator becomes the same polynomial in the arithmetic variable s . The quotient $Q = \mathcal{B}/\Theta V$ records analytic classes. The original two-term coefficient resolution is

$$[V \oplus V \xrightarrow{(\phi, \psi) \mapsto \Theta(\phi - \hat{\psi})} \mathcal{B}].$$

Its chart and tensor-resolution identifications are proved in the original audit; the current arithmetic map uses its top tensor cohomology $Q^{\otimes k}$. [R09 §1; R14 AKP1–2.]

This is a direct debt to Ralf Meyer's spectral construction, not merely a generic similarity to the literature. In Meyer, *A spectral interpretation for the zeros of the Riemann zeta function*, §3, Definition 3.1 and equation 11, $Zf = \sum_{n \geq 1} f(nx)$, the even Schwartz subspace has $f(0) = \widehat{f}(0) = 0$, and the Poisson identity supplies its two asymptotic conditions. Here V is that subspace under the stated Fourier convention and $\Theta = 2Z$. In fact the original quotient Q is topologically and equivariantly Meyer's quotient $\mathcal{H}_-^0$, as the explicit comparison below shows. It should not be presented as a newly invented quotient. The source conventions are related as follows: Meyer's completed zeta denoted ξ does not contain the same $s(s-1)$ factor as the programme's g , and the temporary generator $x\partial_x$ in his Proposition 3.2 has the opposite sign from D . His closed-range result is Theorem 3.3. Meyer in turn explicitly discusses Alain Connes's earlier spectral interpretation. These are specific debts to Meyer and the antecedent he identifies in Connes [Meyer \(2005\)](#); [Connes \(1999\)](#). The version used for the convention comparison is Meyer's pinned arXiv version 3, §§2–4.

For completeness, put $f(t) = F(e^t)$. The source seminorms become $p_{b,j}(F) = \sup_t e^{bt} |f^{(j)}(t)|$. Meyer requires $e^{st}f(t)$ to be Schwartz for each real s . If $b_- < s < b_+$ are integers, every polynomial weight $(1 + |t|)^n$ is bounded by a constant times $e^{(b_+ - s)t}$ on $t \geq 0$, and by a constant times $e^{(b_- - s)t}$ on $t \leq 0$. Leibniz's rule therefore gives

$$\sup_t (1 + |t|)^n |\partial_t^j (e^{st}f)| \leq C \sum_{r=0}^j \binom{j}{r} |s|^{j-r} (p_{b_-,r}(F) + p_{b_+,r}(F)).$$

Conversely $e^{bt}f^{(j)} = (\partial_t - b)^j(e^{bt}f)$, so each $p_{b,j}$ is bounded by a finite sum of Meyer's Schwartz seminorms. The identity is consequently a homeomorphism $\mathcal{B} \simeq \mathcal{H}_-$. Since $2V = V$, its quotient map is the literal identity $Q \rightarrow \mathcal{H}_-, [F] \mapsto [F]$. Both spaces carry $U_a F(x) = F(x/a)$, so this map also intertwines their dilation representations. This exact precedent determines the credit at the first definition of Q .

Now stipulate a complete zero quartet

$$\begin{aligned} \rho &= \tfrac{1}{2} + \delta + i\gamma, & 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad m \geq 1, \\ \mathcal{R} &= \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, & h(s) &= \prod_{\alpha \in \mathcal{R}} (s - \alpha)^m. \end{aligned}$$

The hypothesis specifies that m is the complete order at each member of this packet. Consequently $v_h = g/h$ is entire at those four points and $v_h(\alpha) \neq 0$. The primary algebra $E_h = \mathbb{C}[s]/(h)$ has dimension $4m$. It retains derivatives through order $m-1$, not just the four values. Multiplication by

$$v_h = j_h v_h \in E_h$$

is an automorphism because its constant value in each local primary factor is nonzero. Its matrix in raw derivative coordinates is triangular, with entries

$$\binom{r}{s} v_h^{(r-s)}(\alpha) \quad (r \geq s).$$

The off-diagonal entries matter whenever $m > 1$. Replacing this matrix by four scalar values would change the map. The original Euler-section proof constructs F_h with $\mathcal{M}F_h = v_h$ and an exact jet section $\sigma_h : E_h \hookrightarrow Q$. It proves those maps on their stated domains. [R09 §1; R14 OCS1, OCS12–18, AKP2; full analytic proof in original audit chapter 04.]

4 Why the sum algebra is smaller than the full tensor product

Fix $k = 4l + 1$, $l \geq 1$, and retain

$$c = k/2, \quad e_k = 1 + k(m-1), \quad \lambda_{ab} = c + (2a-k)\delta + i(2b-k)\gamma,$$

$$\chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^{e_k}, \quad q = e_k(k+1)^2, \quad E = \mathbb{C}[S]/(\chi).$$

The full ordered tensor product is $E_h^{\otimes k}$, of dimension $(4m)^k$. The programme does not identify it with the q -dimensional algebra E . It gives the exact injection

$$\eta : E \longrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(s_1 + \cdots + s_k) 1. \quad (\text{A})$$

Here is the reason for the exponent e_k , including its constant. In a fixed ordered primary tensor block write $s_i = \alpha_i + n_i$, where $n_i^m = 0$. The nilpotents commute. Every monomial in $(\sum_i n_i)^{k(m-1)+1}$ has some exponent at least m , so that power vanishes. In the preceding power the coefficient of $n_1^{m-1} \cdots n_k^{m-1}$ is

$$\frac{[k(m-1)]!}{[(m-1)!]^k},$$

which is nonzero over $\mathbb{C}$; this is the only surviving monomial of that total degree. The nilpotence index is therefore exactly e_k . The possible sums of the original roots are precisely the distinct λ_{ab} . Their real and imaginary parts separately distinguish a, b , since $\delta, \gamma > 0$. The annihilator of the cyclic unit for multiplication by $\sum_i s_i$ is thus exactly χ . Polynomial substitution has kernel (χ) , and multiplication by the invertible full unit $v_h^{\otimes k}$ preserves injectivity. This proves (A) and the intertwining identity

$$\eta M_S = \left(\sum_i M_{s_i} \right) \eta.$$

This calculation illustrates the intended role of an exact map: the smaller cyclic module is a specified submodule of the complete tensor object. It does not erase the other tensor directions. [R14 OCS1–3, §1 pp.20–21; CAI10, §198 p.624.]

At the function level the corresponding source is

$$\mathcal{V}_{h,k}(P) = P(D_1 + \cdots + D_k) F_h^{\otimes k}, \quad q^{(k)} \mathcal{V}_{h,k}(P) = \sigma_h^{\otimes k} \eta[P].$$

The relation $P \mapsto P + \chi\kappa$ changes the representative by an explicit coboundary. Ordered monic division gives polynomials $Q_a(\mathbf{s})$ with

$$\chi(s_1 + \cdots + s_k) = \sum_{a=1}^k h(s_a) Q_a(\mathbf{s}).$$

The final remainder vanishes by the annihilator calculation. With the original ϕ_0 satisfying $\Theta\phi_0 = h(D)F_h$, placed in the degree-zero plus leg as $(\phi_0, 0)$, define

$$\Psi_\kappa = \sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa \left(\sum_b D_b \right) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}).$$

In the a -th summand the tensor differential contributes another sign $(-1)^{a-1}$; the two signs multiply to 1. Applying the differential therefore gives exactly $d\Psi_\kappa = \mathcal{V}_{h,k}(\chi\kappa)$. This proves representative independence with its actual homotopy. By contrast, lying in a subsequent observation kernel does not make a class a coboundary: the original injection may still carry it to a nonzero arithmetic class. [R14 AKP2–5, pp.110–111.]

5 What the period observation sees

The additional observation uses the original polynomial primitive $\Phi'_h = h$, $\Phi_h(0) = 0$, a nonzero parameter u , a fixed branch, and the oriented contours

$$\Gamma_j = -\ell_0 + \ell_j, \quad \arg \ell_j = \frac{\arg u + \pi + 2\pi j}{4m+1}.$$

On a degree-less-than- $4m$ representative,

$$(\Pi[P])_j = \int_{\Gamma_j} e^{\Phi_h(z)/u} P(z) dz.$$

The preceding full period theorem supplies convergence and invertibility on its stated domain. In file 03, PDE1–13, the calculation uses the actual determinant differential equation and its Gamma/Fourier base point; this proof is given directly, rather than invoking an unspecified abstract period theorem. It is not permissible to choose period entries independently: each is this integral of the specified representative. Let $n = 4m$, $N = n + 1$, $U = M_{v_h}$ and $\iota[P] = [P(s_1 + \cdots + s_k)]$. The observed map is

$$L = P_k \Pi^{\otimes k} U^{\otimes k} \iota = j\Lambda, \quad P_k = \frac{1}{N} \sum_{a=0}^{N-1} (C^{-a})^{\otimes k}, \quad K = \ker \Lambda,$$

where $\Lambda : E \rightarrow B$ is onto its actual image and j includes that image into the ordered period tensor space. The literal cyclic matrix is

$$(Cx)_j = x_{j+1} - x_1 \ (j < N-1), \quad (Cx)_{N-1} = -x_1.$$

With $\zeta = e^{2\pi i/N}$, $f_t = (\zeta^{jt} - 1)_{j=1}^{N-1}$, one computes $Cf_t = \zeta^t f_t$. The finite average selects the symmetric tensor words whose sum of indices is zero modulo N . However, these Fourier columns have Gram $N(I + \mathbf{1}\mathbf{1}^*)$, so this coordinate diagonalization is not an assertion that the projector is orthogonal in the original metric. The current source retains the induced full tensor Gram and all its cross terms. [R14 OCS2–11, §1 pp.20–23.]

For the simple-quartet case $m = 1$, the corresponding period curve is a quadric with an actual cyclic orbit. Its five-member orbit gives

$$\text{rank } \Lambda = k^2 - 6k + 17, \quad \dim K = 8k - 16.$$

The actual period calculation supplies a radius R_* forcing the relevant orbit on its stated branch domain; the metric applications retain $|u| \geq R_*$, and later pole presentations have their further specified exclusions or sufficient radius. These numbers are not asserted for arbitrary multiplicity. The all- m geometry retains its own Segre–Veronese maps, nilpotents and stabilizer. [R14 OQX3, §3 pp.32–33; PQO/RPC, §§4–5 pp.34–42; OMG, §13 pp.84ff.]

Ranks alone do not determine the desired volumes. The latter depend on the period coefficients and on how their kernel sits inside the original source metric. This is the reason the calculation continues after the exact rank theorem.

6 The arithmetic norm and the two Gamma references

The arithmetic density is

$$w_h(y) = \frac{|v_h(\frac{1}{2} + iy)|^2}{2\pi}, \quad m_{h,k} = w_h^{*k}.$$

For $P \in \mathbb{C}[S]$, the finite source norm is

$$\|P\|_{\text{ar},k}^2 = \int_{\mathbb{R}} |P(c + iy)|^2 m_{h,k}(y) dy.$$

It comes from the same Mellin/theta class; no replacement probability measure is implicit. In particular the full mass is retained. The analytically tractable reference measures are

$$\begin{aligned} \sigma(y) &= \frac{|\Gamma(\frac{1}{4} + iy/2)|^2}{2\pi}, \quad c_\sigma = \sqrt{2\pi}, \quad c_a = 2^{1-2a}\Gamma(2a), \\ r_k^0(y) &= \frac{c_\sigma^k}{c_{k/4}} \frac{|\Gamma(k/4 + iy/2)|^2}{2\pi}. \end{aligned}$$

Their masses are c_σ and c_σ^k , respectively. The monic Gamma-polynomial norm convention is $h_{n,s} = (2\pi)^{s/2} n!(s/2)_n$, with $s = 1$ or k . These are classical Gamma/Meixner–Pollaczek structures, used here with the displayed scaling and mass convention [Koornwinder et al. \(2026\)](#); [Koekoek et al. \(2010\)](#).

An exact identity explains why two source orders can be compared on the same arithmetic quotient. Set

$$b_j = 2j + \frac{1}{2}, \quad f_l(S) = \prod_{j=0}^{l-1} (S - (c + b_j)), \quad C_k = \frac{c_\sigma^k}{c_{k/4}} 4^{-l}.$$

The classical Gamma recurrence [Askey and Roy \(2026\)](#) gives

$$\frac{\Gamma(l + 1/4 + iy/2)}{\Gamma(1/4 + iy/2)} = (i/2)^l \prod_{j=0}^{l-1} (y - ib_j).$$

Taking modulus on the real line gives the complete equality

$$r_k^0(y) = C_k |f_l(c + iy)|^2 \sigma(y). \tag{B}$$

The holomorphic recurrence uses the displayed minus signs before taking the modulus. For example $k = 5$ gives $f_1(S) = S - 3$, $C_5 = 16\pi^2/3$. Since k is odd, each original λ_{ab} has nonzero imaginary part, whereas all roots of f_l are real. Thus f_l and χ are coprime, multiplication $M_f : E \rightarrow E$ is invertible, and

$$\eta M_f = M_{f(\sum_i s_i)} \eta.$$

This supplies the exact arithmetic map behind B . Multiplication by f_l changes the source representative in a known way; it is not silently identified with the identity. [R09 §2, “The exact two-reference multiplication,” and §3, “Joint full jets and the mixed Schur complement.”]

Comparison estimates between these norms and the arithmetic source are proved on *all polynomial directions* before taking restrictions or quotients. If $aH_1 \preceq H_2 \preceq AH_1$ on one coefficient space, then minimizing each inequality over the same affine fibre proves the corresponding quotient inequality. A bound on selected moments or selected monic test vectors alone would not prove that whole-form comparison. The current arithmetic comparison is AMT, and later source-order refinements retain those same fibres and maps. [R14 §8 pp.52ff.; §§192,197,204–211 pp.571ff.,602ff.,648–659.]

7 From a source norm to four determinant returns

For $N \geq q - 1$, the remainder map $J_N : \mathcal{P}_N \rightarrow E$ has kernel $\chi \mathcal{P}_{N-q}$. If H_N is the source Gram in a fixed coefficient frame, its attained quotient Gram is

$$G_N = (J_N H_N^{-1} J_N^*)^{-1}. \quad (\text{C})$$

Indeed $S_N^{\text{rem}} = H_N^{-1} J_N^* G_N$ is a right inverse of J_N . For $z \in \ker J_N$, $z^* H_N S_N^{\text{rem}} = 0$. Every lift of a value is the displayed minimum lift plus such a z , so its squared norm is the minimum squared norm plus $z^* H_N z$. This proves (C), not merely its positivity. In a monic orthogonal-polynomial frame, with $b_n = [p_n]_\chi$ and full squared norm ω_n , the inverse is

$$G_N^{-1} = \sum_{n=0}^N \frac{b_n b_n^*}{\omega_n}.$$

The first q remainder columns form a triangular invertible matrix; hence the inverse exists. [R14 CAI2, §198 p.622.]

Now apply the *actual* observation $\Lambda : E \rightarrow B$ and a fixed inclusion $I : K \hookrightarrow E$. Its kernel metric and attained image metric are

$$\begin{aligned} H_{K,N} &= I^* G_N I, & Q_{B,N} &= (\Lambda G_N^{-1} \Lambda^*)^{-1}, \\ S_N &= G_N^{-1} \Lambda^* Q_{B,N}, & \Lambda S_N &= I_B, & I^* G_N S_N &= 0. \end{aligned}$$

Choose one fixed section S_* of Λ . The frames $[I, S_*]$ and $[I, S_N]$ differ by a triangular matrix with determinant one, since $S_N - S_*$ has image in K . Orthogonality therefore proves

$$\det H_{K,N} \det Q_{B,N} = |\det[I, S_*]|^2 \det G_N. \quad (\text{D})$$

The frame factor is present; it is not set equal to one. Use the original four endpoints

$$N_0 = q - 1, \quad N_1 = q, \quad N_2 = 2q - 1, \quad N_3 = 2q, \quad (\varepsilon_0, \varepsilon_1, \varepsilon_2, \varepsilon_3) = (1, 1, -1, -1).$$

Define the total, kernel and observed-image returns by

$$\mathcal{B}_k = \sum_{i=0}^3 \varepsilon_i \log \det G_{N_i}, \quad F_K = \sum_i \varepsilon_i \log \det H_{K,N_i}, \quad F_B = \sum_i \varepsilon_i \log \det Q_{B,N_i}.$$

Taking this exact signed sum in (D) cancels the fixed factor because $\sum_i \varepsilon_i = 0$, and proves $F_K + F_B = \mathcal{B}_k$. The signs are part of the definition; they are what makes constant frame changes cancel. This same calculation works for the arithmetic and each Gamma source separately. [R14 CAI3–5, pp.622–623; AKP6–10, pp.111–112.]

Schur complements, minimum sections, determinant lemmas and their Woodbury form are classical finite-dimensional linear algebra. The contribution to this programme is the calculation of the original coefficient maps, frames and constants to which those identities are applied, and the proof that they give the original arithmetic/cohomological quantities. The classical linear-algebra identities are credited independently of this programme-specific application [Boyd et al. \(1994\)](#); [Hager \(1989\)](#); [Sherman and Morrison \(1950\)](#); [Woodbury \(1950\)](#).

8 The later native metric and its two correlated row energies

The conductor calculations recast the relevant dual metric in terms of a numerator divided by the actual conductor symbol E_A . Its order $v = \text{ord}_0 E_A$, leading moment $\mu_v \neq 0$ and scalar $\alpha_A = v!/\mu_v$ remain in every division matrix. The metric is the squared norm of that numerator after minimizing over all allowed lower evaluations. Thus its finite raw data consist of a lower-evaluation matrix V and a numerator matrix T , both in the original orthonormal Gamma frame.

At one cutoff set $A = V^*V$, $K_0 = A^{-1}V^*T$. When the next rows v_+, t are appended, the residual row and its conditioning denominator are

$$r = t - v_+ K_0, \quad \lambda = 1 + v_+ A^{-1} v_+^*, \quad h = \lambda^{-1/2} r.$$

If the old attained numerator Gram is D , its new value is exactly

$$D^+ = D + h^* h. \tag{E}$$

To check E , subtract the old attained minimum. A change η in the old lower coefficient costs

$$\eta^* A \eta + \|rx - v_+ \eta\|^2.$$

The normal equations give $\eta = (A + v_+^* v_+)^{-1} v_+^* rx$. Substitution gives $x^* r^* rx / (1 + v_+ A^{-1} v_+^*)$. This proves the update with the complete lower conditioning, rather than minimizing the new row independently of the old rows. [R14 HJR14–15, §73 pp.304–309.]

Let B now denote the actual onto coefficient map in the native quotient and J its kernel inclusion. These local symbols belong to HJR and should not be confused with the earlier period-image space B . Set

$$D_J = J^* D J, \quad D_B = (B D^{-1} B^*)^{-1}, \quad L = D^{-1} B^* D_B.$$

The same attained-section proof gives the exact orthogonal inverse decomposition

$$D^{-1} = J D_J^{-1} J^* + L D_B^{-1} L^*.$$

Consequently define two nonnegative *actual* row energies

$$x = h J D_J^{-1} J^* h^*, \quad y = h L D_B^{-1} L^* h^*.$$

The determinant lemma and one further quotient minimum give

$$\begin{aligned} \log \frac{\det D^+}{\det D} &= \log(1 + x + y), \\ \log \frac{\det D_J^+}{\det D_J} &= \log(1 + x), \quad \log \frac{\det D_B^+}{\det D_B} = \log \left(1 + \frac{y}{1 + x} \right). \end{aligned} \quad (\text{F})$$

For the last identity, a quotient lift of a value a is $La + Jz$ in the stated original coefficient spaces. Its old squared norm is $a^* D_B a + z^* D_J z$, and its additional squared norm is $|hLa + hJz|^2$. Put $t = D_J^{1/2} z$ and $w = hJ D_J^{-1/2}$, so $ww^* = x$. The same one-row normal equation as in (E) gives the minimum additional form $|hLa|^2/(1 + x)$. Its determinant relative to D_B is $1 + y/(1 + x)$, proving the last identity with that denominator. Thus

$$\log(1 + x + y) = \log(1 + x) + \log \left(1 + \frac{y}{1 + x} \right)$$

is the determinant splitting of this precise metric, not a free numerical partition of a known total. [R14 HJR16–18, pp.304–309.]

Summing the rows from $M = q - 1$ through $(2q-1)$ with weight one at those two ends and weight two at every interior row gives

$$S_K = \sum_M w_M \log \left(1 + \frac{y_M}{1 + x_M} \right), \quad S_R = \sum_M w_M \log(1 + x_M), \quad T = S_K + S_R. \quad (\text{G})$$

The original dual/coefficient maps connect S_K to the conductor-transported kernel return; the finite discrepancy from F_K is explicitly controlled. The complete two-window row total satisfies, on the original simple-quartet fixed-period domain,

$$\frac{T}{kq} \longrightarrow 16C_\partial, \quad 21.66851180520 < 16C_\partial < 21.66851180521.$$

The coefficient is the evaluated logarithmic-potential expression

$$C_\partial = 4 \log(4/\pi) - 10 - 2F(2, \pi) + 4\mathcal{L}(2, \pi),$$

where F and $\mathcal{L}$ are the energy and logarithmic moment of the proved equilibrium for $V_{\alpha, \beta}(x) = \beta\sqrt{x} - \alpha \log x$. The exact equilibrium and outward certificate are part of the source. For orientation to the classical logarithmic-potential setting, see [Saff and Totik \(1997\)](#); the specific potential, original arithmetic comparison and receiving constants are calculated in R14. This number is the total coefficient. It does not select the separate leading value of either summand in (G). [R14 HJR19–25 and subsequent receiving refinements; ECL15–18, §59 pp.243–244; current continuation §“Exact working quantities”.]

9 The precise matrix expression that remains

The latest mixed calculation uses the target degree $j = k + 4$, the original source pairs $(1, 1)$ and (k, j) , and eight fixed coefficient subspaces in the four-copy target. In the notation of the current prompt, put

$$I = \text{im}(B_{\text{cof}}J_0), \quad B = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \mathbb{B},$$

$$K_i = I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T_e = S + J.$$

Here F_e , $e = 0, 1$, is the original connecting-map column family at the two low cutoffs. The full connecting constructions and exact coefficient maps are in R14 §§151–159 and §§184–193. The resulting inclusions are the two flags

$$J \subset W_i \subset W \subset T_e, \quad K_i \subset I \subset K_c \subset S.$$

At each original row let D_0 be its preceding positive numerator form, ρ_n its lower-evaluation-corrected row and ζ_n its complete conditioning scalar. With

$$\mathcal{D}_0 = \text{diag}(D_0, D_0, D_0, D_0), \quad \mathcal{U}_n = \text{diag}(a_n, a_n, a_n, a_n), \quad a_n = \frac{\rho_n}{\sqrt{1 + \zeta_n}},$$

the four-by-four compression associated with a fixed independent frame X for one of the eight subspaces is

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*. \quad (\text{H})$$

The empty subspace gives the zero matrix. Formula (H) is unchanged by an invertible change of columns in X . If $Y = \mathcal{U}_n \mathcal{D}_0^{-1/2}$, it is $Y \Pi_X Y^*$, where Π_X is the orthogonal projector onto $\mathcal{D}_0^{1/2} \text{im } X$. Hence inclusion of subspaces gives positive matrix order. Also

$$YY^* = r_n I_4, \quad r_n = \frac{\rho_n \mathcal{D}_0^{-1} \rho_n^*}{1 + \zeta_n}, \quad 0 \preceq K_{X,n} \preceq r_n I_4.$$

The signed threshold profile $\mathcal{P}_{k,e}$ takes the four eigenvalues of each matrix, including zeros, applies $\log \max(1, \lambda)$, and sums over the original rows with their endpoint/interior weights. The positive subspaces are (T_e, K_c, W_i, K_i) ; the negative ones are (W, S, J, I) . The accepted theorem gives

$$|\mathcal{L}_e - \mathcal{P}_{k,e}| \leq \sum_n \vartheta_n \varepsilon(r_n) \leq 16q_j \log 2,$$

$$\varepsilon(r) = \begin{cases} 4 \log(1 + r), & 0 \leq r \leq 1, \\ 4 \log(4r/(1 + r)), & r \geq 1. \end{cases} \quad (\text{I})$$

This error is $o(kq_k)$. It has already been proved for the unchanged original profile. Its proof uses the two nested flags and eigenvalue monotonicity; it assumes no simultaneous diagonalization. [R14 NTE1–7, §199 pp.631–632.]

The map back to the original allocation is equally important:

$$4S_{K,j} = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \quad (\text{J})$$

The terms are attained quotient returns on the original coefficient spaces: V_{rem} is the remaining target quotient by T_e ; V_s is the connecting source form on S/K_c ; V_A is the invariant source form on I/K_i in the target-degree metric; and V_b is the retained boundary return. Each is computed with the original four endpoints and fixed value frame. These positive returns are correlated with $\mathcal{L}_e$, since all use the same original row data. The degree- k proper-source form

has its own centre and comparison map; it is not substituted for V_A . [R14 CPQ1–20, §194 pp.587–591; EFP, §§204–211.]

Thus the remaining target is to evaluate or rigorously constrain the signed profile in (H)–(I) *together with* the complementary returns in (J), through their actual Gamma recurrence, conductor division and coefficient maps. An arbitrary matrix model, a dimension fraction, or the known total T does not calculate this expression. Conversely, the fact that the expression is not yet evaluated is not an obstruction theorem for the entire cohomological programme.

10 Why an allocation result must pass through the whole action

Return to the complete arithmetic quotient E , its multiplication $A = M_S$ and its metric G_N . The monic arithmetic recurrence, in the original coordinate $S = c + iy$, is

$$Sp_n = p_{n+1} + cp_n - \frac{\omega_n}{\omega_{n-1}}p_{n-1}.$$

It gives the rank-two identity

$$AG_N^{-1} + G_N^{-1}A^* - kG_N^{-1} = \frac{b_{N+1}b_N^* + b_Nb_{N+1}^*}{\omega_N}.$$

Reflection $P(S) \mapsto P(k - S)$ is an isometry of the original source and quotient. It forces the total adjacent pairing to vanish. The relative Hermitian action therefore has only two possibly nonzero eigenvalues $\epsilon_N, -\epsilon_N$. In an orthogonal kernel/image split, the individual mixed pairings need not vanish: they cancel in their sum, and the mixed exterior norm retains the term $2|z_K|^2$. [R14 CAI6–9, p.623.]

The exact same-class exterior argument gives

$$\frac{\delta q(k+1)}{2} \leq \epsilon_N \quad (N \geq q-1). \quad (\text{K})$$

This is a lower benchmark from the hypothetical quartet itself. It is not an independent contradiction. To contradict that quartet by this route, a proved upper control would have to conflict with the lower benchmark on the same original object and regime.

Put $d_n = \det G_n / \det G_{n-1}$, so $0 < d_n < 1$ at the relevant rows. The adjacent formulas retain both factors:

$$\begin{aligned} \epsilon_{q-1}^2 &= \frac{\omega_q}{\omega_{q-1}}(d_q^{-1} - 1), \\ \epsilon_N^2 &= \frac{\omega_{N+1}}{\omega_N}(1 - d_N)(d_{N+1}^{-1} - 1) \quad (N \geq q). \end{aligned}$$

After taking the two original windows, with the original weights, the action is exactly

$$\begin{aligned} J_k^{\text{action}} &= \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\ \mathcal{W}_k^{\text{ar}} &= \log \frac{\omega_{2q-1}\omega_{2q}}{\omega_{q-1}\omega_q}. \end{aligned} \quad (\text{L})$$

The two nonnegative penalties are the complete weighted sums of $-\log(1 - d_n)$ specified in CAI13. Formula (L) follows by multiplying the adjacent identities: monic-norm ratios telescope, reciprocal d_n ratios give the four source volumes, and the remaining contraction

factors give precisely those penalties. The original kernel and reference boundary contributions cancel through $F_K + F_B = \mathcal{B}$. The arithmetic volume return $\mathcal{B}_k^{\text{ar}}$ remains in (L); it is not a coordinate constant. [R14 CAI11–15, §198 pp.624–625.]

This explains what a future result can claim. A calculation of the intrinsic mixed allocation is meaningful geometry. Its effect on the original action must be obtained by substitution into (L), including its cancellation and every remaining term. It cannot be promoted to an action upper bound by dropping the term that cancels it.

11 The growing-degree result already proved

The preceding 692-page addition concerns the complete relation polynomial. In the real coordinate define

$$Q(y) = i^{-q} \chi(c + iy), \quad R_k = k \sqrt{\delta^2 + \gamma^2}.$$

Its roots occur in opposite pairs, with their full multiplicities, and lie in the disk of radius R_k . For every polynomial P of degree at most r , compare

$$H_0(P) = \int_{\mathbb{R}} |y|^{2q} |P(y)|^2 d\sigma(y), \quad H_Q(P) = \int_{\mathbb{R}} |Q(y)P(y)|^2 d\sigma(y).$$

The root-sensitive full-form theorem gives explicit two-sided comparison, including the complete inner region on both half-lines. In its exact notation

$$\begin{aligned} \alpha &= \pi/2, \quad s_q = 2q + 1, \quad Y_{q,r} = \frac{s_q^2}{32\alpha(r + s_q)}, \quad \rho_{q,r} = R_k/Y_{q,r}, \\ \eta_{q,r} &= \frac{c_+}{c_-} \left(1 + \frac{4r + 4q + 2}{\alpha} \right) s_q (3/8)^{s_q}, \quad c_- = \sqrt{2}e^{-7/6}, \quad c_+ = \sqrt{2\sqrt{5}}e^{2/3}. \end{aligned}$$

Under its actual finite guards $2R_k \leq Y_{q,r}$, $\eta_{q,r} < 1$, it proves

$$(1 - \rho_{q,r}^2)^q (1 - \eta_{q,r}) H_0(P) \leq H_Q(P) \leq ((1 + \rho_{q,r}^2)^q + \eta_{q,r}) H_0(P). \quad (\text{M})$$

These are proved conditions with a nonempty verified arithmetic asymptotic range, not missing hypotheses inserted for the desired conclusion. The proof controls all directions through classical Laguerre moment/evaluation bounds Szegő (1975); Koornwinder et al. (2026), the paired-root estimate and a direct inner-region calculation. It then takes the original monic minima and the original affine quotient minima. The actual relation sequence remains

$$0 \longrightarrow \mathcal{P}_r \xrightarrow{\times \chi} \mathcal{P}_{q+r} \xrightarrow{J_{q+r}} E \longrightarrow 0.$$

The map $P(S) \mapsto i^{-r} P(c + iy)$ on monic degree- r polynomials retains its phase and inverse; the entire unit in the arithmetic cohomology map is unchanged. [R14 RGC1–8, RGC18–24, §217 pp.685–690; RGI1–3, §215 p.682.]

A separately proved all-degree monic reference bound has signal

$$F_{q,r} = \frac{q^2}{4(q+r) \log(8(q+r)/q)}.$$

Combining that bound with (M) and the full arithmetic source comparison yields canonical allowance lower bounds for $r_k = \lfloor qk^\theta \rfloor$, for each fixed $0 < \theta < 2/3$ at $m = 1$, and $0 < \theta < 4/3$ at each fixed $m \geq 2$. The further logarithmic refinement sets

$$(d, A_m^{\deg}) = \begin{cases} (2, 1), & m = 1, \\ (3, m - 1), & m \geq 2, \end{cases} \quad \theta_* = 2(d - 1)/3,$$

$$r_k = \left\lfloor \frac{qk^{\theta_*}}{(\log k)^\zeta} \right\rfloor, \quad \zeta > 1/3,$$

and proves

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \geq (1 - o(1)) \frac{A_m^{\deg}}{8\theta_*} k^{(d+2)/3} (\log k)^{\zeta-1}. \quad (\text{N})$$

The proof retains the floor, the original $q = e_k(k+1)^2$, the full root radius, and the exact adjacent factors. The new endpoints are $q - 1, q, q + r_k - 1, q + r_k$; the old high endpoints cannot be reused after changing r . The root-error-to-signal calculation is $O((\log k)^{1-3\zeta}) + o(1)$, which explains the strict condition $\zeta > 1/3$. That earlier estimate established neither $\zeta = 1/3$ nor the uncorrected pure-power endpoint; the successor in Section 12 proves a larger range. [R14 RGC16–32, pp.688–692; RGI11–14, §213 pp.673–675; RGI8–10, §198 pp.629–630.]

These canonical lower bounds eventually exceed the benchmark (K) by a ratio tending to infinity in their stated ranges. They locate the behavior of this specified action construction. They do not evaluate the independent mixed profile in (H)–(J), and they do not close the proposed tau/Deligne programme. A further analytic extension must prove its error for these same forms and compute its effect through (L).

12 The accepted moving-cutoff successor

The accepted 726-page edition extends the previous comparison to degrees whose ratio to the relation degree grows. This extension is useful because the original question concerns increasingly large spaces of representatives. The quotient, the complete zero multiplicities, and the arithmetic norm all stay fixed as defined above. The degree of a representative is allowed to increase, and the estimates must control that increase uniformly. Write **R726** for this edition; its complete new proofs are GRF1–27 (pp. 698–705), GRE1–43 (pp. 706–712), GEL1–13 (pp. 713–717), and GRA1–36 (pp. 718–726).

The analytic comparison uses the original Gamma measure and classical Laguerre and elliptic-integral formulas. Their human sources are Szegő (1975), the authors of DLMF Chapter 18 (Koornwinder et al., 2026), and B. C. Carlson’s DLMF Chapter 19 (Carlson, 2026). The parameter-dependent estimates are proved in GRF, GRE and GEL; citing those classical formulas does not attribute the programme’s arithmetic transfer to their authors.

12.1 The scalar ratio measures a constraint on the original source

For an integer $j \geq 0$, let ω_d^{ar} be the least squared arithmetic norm of a monic degree- d polynomial, and put

$$\nu_j^{\text{ar}} = \min_{P \text{ monic, deg } P=j} \|\chi P\|_{\text{ar}}^2, \quad T_j = \nu_j^{\text{ar}} / \omega_{q+j}^{\text{ar}}, \quad \beta_d = \omega_d^{\text{ar}} / \omega_{d-1}^{\text{ar}}.$$

Both minima are on their displayed affine spaces with the original arithmetic norm. Every χ^P in the first minimum is monic of degree $q+j$, but its required divisibility restricts the available representatives. The ratio records the cost of that restriction. The quotient determinant calculation gives

$$d_{q+j} = T_j^{-1}, \quad \epsilon_{q+j-1}^2 = \beta_{q+j}(T_j - 1) \begin{cases} 1, & j = 0, \\ 1 - T_{j-1}^{-1}, & j \geq 1. \end{cases} \quad (\text{O})$$

To see the first identity, use the source frame $(\chi, \chi S, \dots, \chi S^j; 1, S, \dots, S^{q-1})$. Its determinant has modulus one after arranging its monic columns by degree. Orthogonal elimination therefore factors the source Gram determinant into the relation Gram determinant and the quotient determinant. Taking consecutive ratios yields $d_{q+j} = \omega_{q+j}^{\text{ar}} / \nu_j^{\text{ar}}$. Substitution into the adjacent formulas of the preceding section proves the second identity, including the first-row exception. The complete source calculation is R726 GRA9–17, pp. 719–721.

12.2 The larger degree range and its exact scale

Retain $k = 4\ell + 1$, $q = [1 + k(m-1)](k+1)^2$, and the fixed complete quartet. Let r_k be an integer sequence and set

$$t_k = r_k/q \rightarrow \infty, \quad t_k \log t_k = o(q/k), \quad H_k = \frac{q}{t_k \log t_k}.$$

The accepted result, including the whole earlier band rather than just its last row, is

$$\min_{0 \leq j \leq r_k} \log T_j \sim H_k, \quad \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \sim H_k/2. \quad (\text{P})$$

Here $H_k/k \rightarrow \infty$. The moving comparison first bounds the original full forms on every coefficient vector. Taking their minima on the same monic and quotient fibres then proves the scalar transfer. Its moving root error and its arithmetic error are controlled throughout the specified range. Small degrees, compact positive degree ratios, and growing ratios are joined by the complete uniform argument in R726 GRB5–8, pp. 678–679; the scalar providers are GRA18–25. Thus (P) is a theorem of the stated source edition, with an explicit range, not an inference from a numerical fit.

For a concrete consequence, retain the earlier pair $(d, A_m^{\text{deg}}) = (2, 1)$ for $m = 1$ and $(3, m-1)$ for fixed $m \geq 2$. For

$$r_k = \left\lfloor \frac{qk^\theta}{(\log k)^\zeta} \right\rfloor,$$

one obtains

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \sim \frac{A_m^{\text{deg}}}{2\theta} k^{d-\theta} (\log k)^{\zeta-1} \quad (\text{Q})$$

when $0 < \theta < d-1$, for any fixed real ζ , or when $\theta = d-1$ and $\zeta > 1$. Indeed the floor gives $t_k = k^\theta (\log k)^{-\zeta} + O(1/q)$, whence

$$\frac{t_k \log t_k}{q/k} \sim \frac{\theta}{A_m^{\text{deg}}} k^{\theta-d+1} (\log k)^{1-\zeta}.$$

This tends to zero in exactly the displayed cases, and substitution into (P) proves (Q). The earlier pure-power ranges $\theta < 2/3$ and $\theta < 4/3$ have therefore been enlarged to $\theta < 1$ and $\theta < 2$, respectively, with the stated logarithmic approach to the new endpoints.

12.3 A separately proved enclosure at the critical coefficient

The equality case $\theta = d - 1, \zeta = 1$ needs its actual arithmetic error. Define from the original density

$$\vartheta_h = \int_{-1}^1 w_h(y) dy, \quad M_{\pi/2} = \int_{\mathbb{R}} e^{\pi y/2} w_h(y) dy, \quad \mathfrak{s}_h = \frac{\pi}{2} + \log \frac{M_{\pi/2}}{\vartheta_h}.$$

The proved tilted tail bound makes $M_{\pi/2}$ finite. Evenness gives

$$M_{\pi/2} = \int \cosh(\pi y/2) w_h(y) dy \geq \int w_h \geq \vartheta_h > 0,$$

so $\mathfrak{s}_h \geq \pi/2$. For a fixed coefficient satisfying

$$0 < c_{\deg} < \frac{A_m^{\deg}}{2(d-1)\mathfrak{s}_h}, \quad r_k = \left\lfloor \frac{c_{\deg} q k^{d-1}}{\log k} \right\rfloor, \quad B = \frac{A_m^{\deg}}{c_{\deg}(d-1)},$$

R726 GRB11–13, pp. 679–680, proves

$$\frac{B}{2} - \mathfrak{s}_h \leq \liminf \frac{1}{k} \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \leq \limsup \frac{1}{k} \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \leq \frac{B}{2} + \mathfrak{s}_h. \quad (\text{R})$$

All limits use the original subsequence $k = 4\ell + 1$. The lower endpoint is positive. This is an enclosure; the earlier asymptotic equivalent (P) is not asserted at this boundary. The coefficient $c_{\deg}$ is independent of k and is distinct from the unchanged polynomial centre $c = k/2$.

12.4 The complete shifted action keeps every endpoint

For an integer shift $s \geq 1$, put $a = q + s - 1$, $b = a + q$, and $w_0 = w_q = 1$, $w_i = 2$ for $1 \leq i < q$. Then

$$J_{k,s}^{\text{action}} = 2 \sum_{i=0}^q w_i \log \epsilon_{a+i} = \log \frac{V_a V_{a+1}}{V_b V_{b+1}} + \log \frac{\omega_b \omega_{b+1}}{\omega_a \omega_{a+1}} - \mathcal{P}_{k,s}, \quad (\text{S})$$

where $V_N = \det G_N$, $\omega_N = \omega_N^{\text{ar}}$, and

$$\mathcal{P}_{k,s} = \sum_{j=0}^{q+1} c_j [-\log(1 - T_{s-1+j}^{-1})], \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1).$$

For any positive sequence z_N , collecting coefficients gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Apply this to V and ω in (O). The two adjacent contractions give $c_j = w_j + w_{j-1}$, with absent weights zero, and therefore total penalty weight $4q$. This proves (S). R726 GRA27–36 and GRB9–10 supply all finite error bounds and the transfer through the original kernel inclusion and quotient map.

For $t = s/q \rightarrow \infty$ with $t \log t = o(q/k)$, those bounds give

$$J_{k,s}^{\text{action}} \sim \frac{2q^2}{t \log t}, \quad \log \mathcal{P}_{k,s} \sim -\frac{q}{t \log t}.$$

At the critical choice $s = r_k$ in (R), the separate enclosure is

$$2(B - \mathfrak{s}_h) \leq \liminf \frac{J_{k,s}^{\text{action}}}{qk} \leq \limsup \frac{J_{k,s}^{\text{action}}}{qk} \leq 2(B + \mathfrak{s}_h).$$

These advances concern the complete correlated action. The intrinsic signed eight-matrix allocation and each CPQ complementary return still require their own calculation on the maps already defined above.

13 A usable reading route and credit map

The following route leads from the original objects to their exact maps, the current estimates and the remaining calculation:

Question	Exact proof location in the fixed edition
What are the quartet, multiplicities and original tensor map?	R09 §1; R14 OCS1–3, §1 pp.20–21; CAI1 and CAI10, §198 pp.622,624.
What is the actual period/cyclic observation?	R14 OCS2–21, §1 pp.20–25; OQX3 §3 pp.32–33; PQO/RPC §§4–5 pp.34–42.
How is a representative change proved to preserve the cohomology class?	R14 AKP1–5, §26 pp.110–111; full analytic and categorical foundations in Appendix A.
Where are all full-unit and confluent-jet factors retained?	R09 §2; R14 OCS12–18, §1 pp.23–25; RGI1–3 §215 p.682.
How are arithmetic and Gamma metrics related?	R09 §§2–3; R14 AMT §8 pp.52ff.; NSO/EOR §197 pp.602–620; EFP §§204–211 pp.648–659.
Where are the actual minimum metrics, frames and four signs proved?	R14 AKP6–10 §26 pp.111–112; CAI2–5 §198 pp.622–623; HJR14–22 §73 pp.304–309.
What total is known, and why does it not give the two allocations?	R14 ECL15–18 §59 pp.243–244; HJR16–25 §73 pp.304–309; current prompt, “Exact working quantities.”
Where is the mixed profile connected to the original complex?	R14 §§151–159 pp.466–487; CPR/BTK/RPI/RRC §§200–203 pp.633–647; CPQ §194 pp.587–591.
What exactly remains to estimate?	R14 SFB §195 pp.592–598, NTE1–7 §199 pp.631–632, EFP §§204–211; equations (H)–(J) here.
Why can an allocation bound cancel in the action?	R14 CAI11–15 §198 pp.624–625; changed-window RGI8–10 pp.629–630.
Where are the preceding growing-degree estimates?	R14 RGC1–32 §217 pp.685–692 and RGI1–14.
What is the accepted moving-cutoff advance?	R726 GRF/GRE/GEL/GRA pp.698–726 and GRB1–13 pp.631–632,674–675,678–680; Section 12 here.

A Foundational maps: support, the theta complex, and full arithmetic jets

This appendix supplies the elementary constructions underlying the mathematical route. The order matters: first specify what the scalar and support labels mean, then construct the

cochain complex, then connect its quotient to the actual completed zeta function and its full local jets. The distinction between an established human construction and a calculation made in this programme is maintained throughout. In particular, the theta quotient belongs to the spectral framework developed by Ralf Meyer, following Alain Connes; the mathematical route gives its exact comparison with Meyer's quotient, including the topologies and the factor of two (Meyer, 2005, §§2–4). The language of blueprints used for the structural base is due to Oliver Lorscheid Lorscheid (2012).

A.1 The scalar object and its structural base

The project origin of this construction is documented in ChatGPT 5.4 and Gemini 2.5 DeepThink (2026); The Clankers (2026), work developed under KokunoYumeto's direction with AI assistance. The deposited creator fields are retained in the bibliography. Semiring background is credited to Jonathan S. Golan Golan (1999); the monoid-scheme and blueprint frameworks are credited separately to Anton Deitmar and Oliver Lorscheid Deitmar (2005); Lorscheid (2012). No historical priority claim for the external-zero construction is made.

Let R be a nonzero commutative ring with identity. Put

$$D_R = G(R) = \{\tau\} \sqcup \{r^\bullet : r \in R\}, \quad e_R = 0_R^\bullet.$$

The two displayed symbols D_R and $G(R)$ denote the same object. Its operations are, for $r, s \in R$,

$$r^\bullet + s^\bullet = (r + s)^\bullet, \quad r^\bullet s^\bullet = (rs)^\bullet, \quad \tau + a = a + \tau = a, \quad \tau a = a\tau = \tau.$$

The additive identity is τ , and the multiplicative identity is $1_R^\bullet$. Associativity and commutativity reduce to the ring laws when all arguments are supported; if an argument of a product is τ , that product is τ . For distributivity, when the multiplier is τ both sides are τ . When the multiplier is supported and both summands are supported, distributivity is the ring identity. If a summand is τ , the required identity is the additive identity law. This checks all semiring axioms, including every case involving absence.

The represented zero and the absent scalar obey

$$e_R^2 = e_R, \quad e_R + e_R = e_R, \quad 1_R^\bullet + e_R = 1_R^\bullet, \quad e_R r^\bullet = e_R, \quad e_R \neq \tau.$$

Let $\mathbb{B} = \{0, 1\}$ have addition $1 + 1 = 1$ and its usual multiplication. The scalar and support observations are

$$\begin{aligned} p_R : D_R &\longrightarrow R, & p_R(\tau) &= 0_R, & p_R(r^\bullet) &= r, \\ b_R : D_R &\longrightarrow \mathbb{B}, & b_R(\tau) &= 0, & b_R(r^\bullet) &= 1. \end{aligned}$$

Both preserve addition and multiplication by the preceding case distinction. The map $(p_R, b_R) : D_R \rightarrow R \times \mathbb{B}$ is injective: the second coordinate determines presence, and the first coordinate then determines the present scalar. In particular, the scalar observation alone forgets information that the pair retains.

The structural pointed monoid is

$$\mathbf{F}_{1,\tau} = \{\tau_0, 1\},$$

with 1 an identity and τ_0 absorbing. Its preaddition is the relation that the empty sum represents τ_0 , together with the relations required by the preaddition axioms. No equation $1 + 1 = 1$ is imposed. Let B_R be the blueprint associated with D_R : its pointed multiplicative

monoid is the multiplicative monoid of D_R , and its additive relations are the equalities of finite sums in D_R . Thus the empty sum represents τ , while $r^\bullet + (-r)^\bullet$ represents e_R . The map

$$j_R : \mathbf{F}_{1,\tau} \longrightarrow B_R, \quad j_R(\tau_0) = \tau, \quad j_R(1) = 1_R^\bullet$$

preserves multiplication, the distinguished elements, and every generating preaddition relation. It is therefore a blueprint morphism. Every proper prime ideal contains τ and excludes $1_R^\bullet$; its inverse image under j_R is exactly $\{\tau_0\}$. This gives the structural map to the one-point spectrum of $\mathbf{F}_{1,\tau}$.

There is also a separate localization

$$D_R[e_R^{-1}] \cong \mathbb{B}.$$

Indeed, invertibility and idempotence force $e_R = 1$ after localization. The identity $e_R r^\bullet = e_R$ then identifies every supported scalar with 1. The homomorphism b_R sends e_R to 1, so factors through this localization. Conversely the homomorphism from $\mathbb{B}$ sending 0 to the localized τ and 1 to its unit is well-defined because the localized unit is additively idempotent. The two maps are inverse on their displayed generators. This calculation relates the support observation to the scalar object; it does not replace the structural morphism j_R .

For the original unital inclusion $j : \mathbb{Z} \hookrightarrow \mathbb{C}$, the split lift sends τ to τ and $n^\bullet$ to $j(n)^\bullet$. Direct evaluation on these two cases proves

$$\begin{array}{ccc} D_{\mathbb{Z}} & \xrightarrow{D_j} & D_{\mathbb{C}} \\ p_{\mathbb{Z}} \downarrow & & \downarrow p_{\mathbb{C}} \\ \mathbb{Z} & \xrightarrow{j} & \mathbb{C}. \end{array}$$

The target of the left scalar observation remains the infinite ring $\mathbb{Z}$.

A.2 Reconstructing all support fibres

The project formulation is given in (The Clankers, 2026, Definition 3.5 and Theorems 3.6–3.7). Its transport-to-the-join addition is the Płonka-sum operation on the reduct with addition and supported scalar operations (Płonka, 1967; Bonzio, 2018); the global zero and its scalar action are specified explicitly below. The underlying semilattice decomposition belongs to the tradition of Clifford’s semilattices of groups Clifford (1941); inverse-semimodule characterizations are also treated by Sen, Bhuniya and Maity Sen et al. (2021). The source guide distinguishes that authenticated antecedent from a complete comparison of its hypotheses. Here all maps for the stated scalar object are proved explicitly.

A diagram consists of a join-semilattice L with bottom $\perp$, an R -module V_l for every $l \in L$, and linear maps $\rho_{lk} : V_l \rightarrow V_k$ for $l \leq k$, satisfying $\rho_{ll} = 1$ and $\rho_{kj}\rho_{lk} = \rho_{lj}$. A morphism $(L, V) \rightarrow (L', W)$ is a bottom- and join-preserving map $\alpha : L \rightarrow L'$, together with linear maps

$$f_l : V_l \longrightarrow W_{\alpha(l)}, \quad f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l.$$

Define its reconstructed carrier and operations by

$$\mathcal{T}(L, V) = \coprod_{l \in L} \{l\} \times V_l,$$

$$(l, x) + (k, y) = (l \vee k, \rho_{l, l \vee k} x + \rho_{k, l \vee k} y), \quad \tau(l, x) = (\perp, 0), \quad r^\bullet(l, x) = (l, rx).$$

Its global zero is $(\perp, 0)$. No condition $V_\perp = 0$ is imposed. Threefold addition transports both parenthesizations to $l \vee k \vee j$, where their amplitudes are $\rho_l x + \rho_k y + \rho_j z$. This proves

associativity. Commutativity and the bottom identity follow from the same transport formulas. Supported scalar laws follow from the module laws in each fibre. In particular $e_R(l, x) = (l, 0)$. Distributivity over vectors follows from linearity of the transition maps. Distributivity when a scalar is absent uses the global bottom zero; all remaining cases reduce to the ring laws. Hence $\mathcal{T}(L, V)$ is a D_R -semimodule. The formula

$$\mathcal{T}(\alpha, f)(l, x) = (\alpha(l), f_l x)$$

is linear: the displayed compatibility of the f_l proves its addition law after transport to the join, and the scalar laws hold in the indicated fibre.

Conversely, let M be a D_R -semimodule. Define

$$L_M = e_R M, \quad l \vee k = l + k, \quad \perp = 0_M,$$

$$V_l = \{m \in M : e_R m = l\}, \quad \rho_{lk}(m) = m + k \quad \text{when } l + k = k.$$

The equality $e_R^2 = e_R$ makes every $l \in L_M$ fixed by e_R . The equality $e_R + e_R = e_R$ makes $l + l = l$. Thus addition in L_M is a commutative idempotent monoid, giving the stated join order and bottom. If $m, n \in V_l$, then $e_R(m + n) = l + l = l$, so this fibre is closed under addition. Moreover

$$m + l = (1_R^\bullet + e_R)m = m, \quad m + (-1_R)^\bullet m = e_R m = l.$$

Thus V_l , with zero l , is an abelian group. For a supported scalar, $e_R(r^\bullet m) = l$; the inherited ring-module laws, including $0_R m = l$, make it an R -module. If $l \leq k$, then $e_R(m + k) = l + k = k$. The map $m \mapsto m + k$ sends zero to zero and is additive because $k + k = k$. It is R -linear because $r^\bullet k = k$. Its identity and composition laws follow from $m + l = m$ and $k + j = j$.

A linear map $F : M \rightarrow N$ sends $e_R M$ to $e_R N$ and induces $\alpha(l) = F(l)$. This preserves bottom and joins and restricts to linear maps of the displayed fibres. Naturality is the equality $F(m + k) = F(m) + F(k)$. These constructions are inverse functors up to the following explicit isomorphisms:

$$\mathcal{T}(L_M, V) \longrightarrow M, \quad (l, m) \longmapsto m, \quad M \longrightarrow \mathcal{T}(L_M, V), \quad m \longmapsto (e_R m, m),$$

$$L \longrightarrow e_R \mathcal{T}(L, V), \quad l \longmapsto (l, 0), \quad V_l \longrightarrow \{l\} \times V_l, \quad x \longmapsto (l, x).$$

The first pair are inverse as functions. Their addition law reduces to

$$(m + (l \vee k)) + (n + (l \vee k)) = m + n + (l \vee k) = m + n,$$

since $e_R(m + n) = l \vee k$; their scalar laws follow directly. The second pair preserve the join and every module and transition map by their definitions. All formulas commute with F or (α, f) , proving naturality and hence the asserted equivalence.

For later use, a split-linear map $F = \mathcal{T}(\alpha, f)$ has two precisely related zero conditions:

$$F(l, x) = 0_N \iff \alpha(l) = \perp' \text{ and } f_l x = 0, \quad F(l, x) \in e_R N \iff f_l x = 0.$$

If $L_0 = \alpha^{-1}(\perp')$, their carriers are respectively

$$\mathcal{T}(L_0, (\ker f_l)_{l \in L_0}) \longrightarrow \mathcal{T}(L, (\ker f_l)_{l \in L}),$$

where the arrow is inclusion. The set L_0 is closed under joins. The transition maps preserve these kernels because $f_k \rho_{lk} = \rho'_{\alpha(l), \alpha(k)} f_l$. Any linear map satisfying either zero condition therefore factors uniquely through the corresponding inclusion. The second carrier is the fibre product $M \times_N e_R N$. This proves the universal properties as well as the exact comparison between them.

A.3 The four-point chart and its derived global sections

Let P be the poset

$$+ < \eta < \sigma, \quad - < \eta < \sigma,$$

where $+$ and $-$ are incomparable. Give it the upper-set topology and work in the algebraic coefficient category $\mathcal{A} = \text{Fun}(P, \text{Vect}_{\mathbb{C}})$ of covariant diagrams. For a point x , let $P_x(y) = \mathbb{C}$ when $x \leq y$, and zero otherwise, with identity maps on nonzero arrows. A natural transformation $P_x \rightarrow F$ is determined by the image of 1 at x , and at $y \geq x$ that image must be $r_{xy}v$. Hence

$$\text{Hom}_{\mathcal{A}}(P_x, F) \cong F_x.$$

Evaluation is exact, since diagram kernels and cokernels are computed at each point. Thus P_x is projective.

Let $\underline{\mathbb{C}}$ be the constant diagram. The sequence

$$0 \longrightarrow P_{\eta} \xrightarrow{v \mapsto (v, -v)} P_+ \oplus P_- \xrightarrow{(u, v) \mapsto u+v} \underline{\mathbb{C}} \longrightarrow 0$$

is exact. At $+$ or $-$, its nonzero part is $\mathbb{C} \xrightarrow{1} \mathbb{C}$. At η and σ , its nonzero part is $\mathbb{C} \xrightarrow{(1, -1)} \mathbb{C}^2 \xrightarrow{(1, 1)} \mathbb{C}$; the kernel of addition is the displayed diagonal with opposite signs. All restriction squares commute with the component maps, proving exactness as diagrams.

Writing $r_{\pm} : F_{\pm} \rightarrow F_{\eta}$, compatible global sections are pairs (u, v) with $r_+u = r_-v$; their values at η and σ are forced. Thus $\Gamma(P, F) = \text{Hom}_{\mathcal{A}}(\underline{\mathbb{C}}, F)$, and the projective resolution gives

$$A_{\tau}(F) = R\Gamma(P, F) \simeq C_F, \quad C_F^0 = F_+ \oplus F_-, \quad C_F^1 = F_{\eta}, \quad d_F(u, v) = r_+u - r_-v.$$

The cohomological degrees and sign in this formula are fixed. With $r = r_{\eta\sigma}$, the two maps $C_F \rightarrow F_{\sigma}[0]$ have degree-zero parts

$$c_+(u, v) = rr_+u, \quad c_-(u, v) = rr_-v.$$

Both are chain maps. Their difference is the chain homotopy $h^1 = r$, since $c_+ - c_- = h^1 d_F$, and all degree-one target components vanish. They therefore represent a single natural derived restriction morphism.

A.4 The algebraic right adjoint, including the signs

For a vector space W , define $I_x(W)_y = W$ if $y \leq x$ and zero otherwise. A morphism $F \rightarrow I_x(W)$ is uniquely determined by $\ell : F_x \rightarrow W$: at $y \leq x$ it is ℓr_{yx} , and elsewhere it is zero. Hence

$$\text{Hom}_{\mathcal{A}}(F, I_x(W)) \cong \text{Hom}_{\mathbb{C}}(F_x, W).$$

Every vector space is injective as a $\mathbb{C}$ -module: extend a basis of a subspace to a basis and choose arbitrary images for the extra basis vectors. Since evaluation is exact, $I_x(W)$ is injective in $\mathcal{A}$.

Put

$$K_{\tau}(W)^{-1} = I_{\eta}(W), \quad K_{\tau}(W)^0 = I_+(W) \oplus I_-(W), \quad d_K = (+\text{res}, -\text{res}).$$

It is a bounded complex of injectives. Its Hom complex from a coefficient sheaf F is

$$\left[\text{Hom}_{\mathbb{C}}(F_{\eta}, W) \xrightarrow{\ell \mapsto (\ell r_+, -\ell r_-)} \text{Hom}_{\mathbb{C}}(F_+, W) \oplus \text{Hom}_{\mathbb{C}}(F_-, W) \right]$$

in degrees $-1, 0$. In the convention

$$d_{\text{Hom}}\alpha = d_W\alpha - (-1)^{|\alpha|}\alpha d_F,$$

the complex $\text{Hom}^\bullet(C_F, W[0])$ has exactly these terms and this differential. Consequently there is a natural chain isomorphism

$$R\text{Hom}_{\mathcal{A}}(F, K_\tau(W)) \cong R\text{Hom}_{\mathbb{C}}(\mathbf{A}_\tau(F), W).$$

For a bounded complex $W^\bullet$, an explicit extension is

$$K_\tau(W)^n = I_\eta(W^{n+1}) \oplus I_+(W^n) \oplus I_-(W^n),$$

$$d(s, u, v) = (-d_W s, \text{res}_+ s + d_W u, -\text{res}_- s + d_W v).$$

Its square is zero: the first component is $d_W^2 s$; in each other component the two restriction-of- $d_W s$ terms cancel, leaving $d_W^2 u$ or $d_W^2 v$. For a sheaf F , the degree- n Hom comparison sends

$$(s, u, v) \mapsto (a = (u, v), b = (-1)^{n+1}s) \in \text{Hom}^n(C_F, W).$$

Indeed its differential on C_F^0 is

$$d_W a - (-1)^n b d_F = d_W a + s d_F,$$

which gives the two restriction terms above, and its differential on C_F^1 is $(-1)^{n+1}d_W s$, equal to the image of $-d_W s$ under the degree- $n+1$ comparison. This verifies the phase explicitly. The same identification for bounded coefficient complexes F is obtained by finite totalization with this Hom differential; the component identities just checked are its horizontal identities, while naturality makes it commute with the differential of F .

At $+$, $-$, η , and σ , the stalk complexes for $W[0]$ are respectively

$$[W \xrightarrow{1} W], \quad [W \xrightarrow{-1} W], \quad W[1], \quad 0.$$

Let $S_\eta W$ be the diagram supported only at η . The map $S_\eta W[1] \rightarrow K_\tau(W)$, identity at η and zero at other stalks, is natural and a quasi-isomorphism: the first two displayed stalk complexes are acyclic, and the remaining stalk comparisons are identities or zero-to-zero. Thus a zero stalk at σ does not make this right adjoint the zero object. All assertions here concern the stated algebraic diagram category. This calculation alone asserts neither a continuous-dual identification for a Schwartz quotient nor an involutive Verdier duality on an arithmetic scheme.

A.5 The original analytic source and the two-leg complex

The original Fourier convention and source are

$$\widehat{\phi}(\xi) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i x \xi} dx, \quad V = \{\phi \in \mathcal{S}(\mathbb{R}) : \phi(-x) = \phi(x), \phi(0) = \widehat{\phi}(0) = 0\}.$$

On the positive half-line retain

$$D = -x\partial_x, \quad \mathcal{B} = \left\{ F \in C^\infty(\mathbb{R}_{>0}) : p_{b,j}(F) = \sup_{x>0} x^b |D^j F(x)| < \infty \ (b \in \mathbb{Z}, j \geq 0) \right\},$$

$$\Theta\phi(x) = \sum_{n \in \mathbb{Z} \setminus \{0\}} \phi(nx) = 2 \sum_{n \geq 1} \phi(nx), \quad JF(x) = x^{-1}F(1/x).$$

Poisson summation is the classical input recorded also in Meyer's construction (Meyer, 2005, §3, equation (11)). In this convention it follows by periodizing a Schwartz function: $\sum_{n \in \mathbb{Z}} \phi(y + nx)$ is a smooth x -periodic function whose k -th Fourier coefficient is $x^{-1} \hat{\phi}(k/x)$, by a change of variables on each interval of the periodization. The original series and all its derivatives converge uniformly on a period by Schwartz decay; the Fourier coefficients decrease faster than every power, by repeated integration by parts. Its Fourier series therefore converges absolutely to the periodized function. Evaluation at $y = 0$ gives

$$\sum_{n \in \mathbb{Z}} \phi(nx) = x^{-1} \sum_{k \in \mathbb{Z}} \hat{\phi}(k/x).$$

For $\phi \in V$, removal of both zero terms yields

$$\Theta\phi(x) = x^{-1} \Theta \hat{\phi}(1/x), \quad J\Theta = \Theta\mathcal{F}.$$

Fourier inversion gives $\mathcal{F}^2\phi(x) = \phi(-x)$; hence $\mathcal{F}^2 = 1$ on even functions, and it exchanges the two vanishing moments. Thus it preserves V .

Here are the actual estimates ensuring that the theta operator has the stated target. For $N > 1$, $x \geq 1$, and $j \geq 0$,

$$|D^j \Theta\phi(x)| \leq 2\zeta(N) x^{-N} \sup_{y>0} y^N |D^j \phi(y)|.$$

Each sum converges normally on compact positive intervals. The identity $DJ = J(1 - D)$, checked by differentiation, and the preceding Poisson identity give at $0 < x \leq 1$ a finite sum of the same bounds for $\hat{\phi}$ at $1/x$. For every desired b, j , choose N large enough that $N + b - 1 \geq 0$ for the latter bounds and $N - b \geq 0$ for the former. They bound $p_{b,j}(\Theta\phi)$ by finitely many Schwartz seminorms of ϕ and $\hat{\phi}$. The Fourier transform is continuous on the Schwartz space, as follows by differentiating its integral and integrating by parts. This proves continuity $V \rightarrow \mathcal{B}$.

The map Θ is injective. For fixed $x > 0$, choosing a Schwartz exponent $N > 2$ makes $\sum_{n,r \geq 1} |\phi(nrx)|$ converge: it is bounded by a constant times $\sum_{n,r} (nr)^{-N}$. The divisor identity for the Möbius function, obtained by expanding $\prod_{p|d} (1 - 1)$, therefore gives

$$\frac{1}{2} \sum_{n \geq 1} \mu(n) \Theta\phi(nx) = \sum_{d \geq 1} \left(\sum_{n|d} \mu(n) \right) \phi(dx) = \phi(x).$$

Evenness and the prescribed value at zero then recover ϕ on the full real line.

For the diagram $\mathcal{T} = (V, V, \mathcal{B}, 0)$ with restrictions Θ and $J\Theta$, the chart resolution is the literal complex

$$C_\theta = [V \oplus V \xrightarrow{d} \mathcal{B}], \quad d(\phi, \psi) = \Theta(\phi - \hat{\psi}),$$

in degrees 0, 1. Its image is ΘV , since Fourier preserves V and every $\Theta\phi$ is $d(\phi, 0)$. Injectivity of Θ consequently proves

$$H^0(C_\theta) = \{(\hat{\psi}, \psi) : \psi \in V\}, \quad H^1(C_\theta) = Q = \mathcal{B}/\Theta V.$$

For a singleton mask the differential is respectively Θ or $-\Theta\mathcal{F}$. Both are injective and have image ΘV , giving zero degree-zero cohomology and the same degree-one quotient Q . Coordinate inclusion of that leaf and the identity on $\mathcal{B}$ form a chain map to the two-leg complex; on H^1 it is the identity of Q . Reconstructing the full support diagram gives

$$\tilde{H}_\tau^1 = \{\tau\} \sqcup \coprod_{\emptyset \neq A \subseteq \{+, -\}} \{A\} \times Q.$$

The skeleton map is $\tau \mapsto \tau$, $(A, [F]) \mapsto (A, 0)$. Its inverse image of the global absent zero is $\{\tau\}$; its inverse image of the whole supported skeleton is the entire carrier. These assertions follow from the two zero conditions already proved in Section A.2.

A.6 The Gaussian, its Mellin transform, and every original constant

Put

$$u(x) = e^{-\pi x^2}, \quad \phi_*(x) = (4\pi^2 x^4 - 6\pi x^2)e^{-\pi x^2}.$$

The positive integral $I = \int_{\mathbb{R}} u(x) dx$ satisfies

$$I^2 = \int_{\mathbb{R}^2} e^{-\pi(x^2+y^2)} dx dy = 2\pi \int_0^\infty e^{-\pi r^2} r dr = 1,$$

so $I = 1$. Differentiation under the Fourier integral and integration by parts, justified by Gaussian decay, give $\partial_\xi \widehat{u} = -2\pi i \xi \widehat{u}$. Its initial value is $\widehat{u}(0) = 1$, hence $\widehat{u}(\xi) = e^{-\pi \xi^2}$. The same operations for a Schwartz function give

$$\mathcal{F}(xf) = \frac{i}{2\pi} \partial_\xi \widehat{f}, \quad \mathcal{F}(f') = 2\pi i \xi \widehat{f}, \quad \mathcal{F}D = (1 - D)\mathcal{F}.$$

Direct differentiation gives

$$Du = 2\pi x^2 u, \quad D^2 u = (4\pi^2 x^4 - 4\pi x^2)u, \quad \phi_* = D(D - 1)u.$$

Consequently

$$\widehat{\phi_*} = (1 - D)(-D)u = D(D - 1)u = \phi_*.$$

This proves that ϕ_* is even Schwartz, $\phi_*(0) = 0$, and $\widehat{\phi_*}(0) = 0$, so $\phi_* \in V$. The operator D preserves V : it preserves Schwartz regularity and evenness, has value zero at zero, and integration by parts gives $\int_{\mathbb{R}} D\phi = \int_{\mathbb{R}} \phi = 0$ when $\phi \in V$. Hence every polynomial in D used below remains on the original source.

For $F \in \mathcal{B}$, define

$$\mathcal{M}F(s) = \int_0^\infty F(x) x^s \frac{dx}{x}.$$

On every compact set of s , choose integers strictly below and strictly above all occurring real parts. The two corresponding seminorms bound the integral at zero and infinity, including every power of $\log x$. Differentiation under the integral therefore proves that $\mathcal{M}F$ is entire. Integration by parts, with vanishing endpoint term, gives

$$\mathcal{M}(DF)(s) = s \mathcal{M}F(s).$$

For the positive-half-line transform $M_+ \phi_*$, the substitution $t = \pi x^2$ gives, initially for $\Re s > 0$,

$$\begin{aligned} M_+ \phi_*(s) &= \pi^{-s/2} \left(2\Gamma\left(\frac{s}{2} + 2\right) - 3\Gamma\left(\frac{s}{2} + 1\right) \right) \\ &= \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2). \end{aligned}$$

The recurrence used here is obtained by integration by parts in the defining Gamma integral. For $\Re s > 1$, absolute convergence permits summation inside the theta integral, since

$$\sum_{n \neq 0} \int_0^\infty |\phi_*(nx)| x^{\Re s} \frac{dx}{x} = 2\zeta(\Re s) \int_0^\infty |\phi_*(y)| y^{\Re s} \frac{dy}{y} < \infty.$$

Therefore, with $f_0 = \Theta \phi_*$,

$$\mathcal{M}f_0(s) = 2\zeta(s) M_+ \phi_*(s) = s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s) = g(s) = 2\xi(s).$$

Here $\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s)$ is the programme's Riemann convention. The factor two from the integer sum and the factor one half in $M_+ \phi_*$ both remain. The entire function

$\mathcal{M}f_0$ provides the continuation in this identity. Also $Jf_0 = f_0$, so the substitution $x \mapsto 1/x$ proves $g(1-s) = g(s)$. Reality of f_0 gives $\overline{g(\bar{s})} = g(s)$.

For an arbitrary $\phi \in V$, the same absolutely convergent calculation gives

$$\mathcal{M}\Theta\phi(s) = 2\zeta(s)M_+\phi(s).$$

Since evenness and $\phi(0) = 0$ imply $\phi(x) = O(x^2)$ at zero, $M_+\phi$ is holomorphic for $\Re s > -2$. In particular, on the strip $0 < \Re s < 1$ set

$$H_\phi(s) = \frac{2\pi^{s/2}M_+\phi(s)}{s(s-1)\Gamma(s/2)}.$$

It is holomorphic there and continuation of the preceding identity gives

$$\mathcal{M}\Theta\phi = gH_\phi, \quad H_{\phi_*} = 1, \quad H_{P(D)\phi_*}(s) = P(s).$$

For the last identity, integration by parts on the positive half-line gives $M_+(D\phi) = sM_+\phi$ on this strip; its endpoint term vanishes.

A.7 Euler inversion and a source section for all local jets

Let Z be a nonempty finite set of actual zeros of g in $0 < \Re s < 1$, each retained with its full order m_ρ . Define

$$h(X) = \prod_{\rho \in Z} (X - \rho)^{m_\rho}, \quad d_h = \deg h, \quad E_h = \mathbb{C}[X]/(h), \quad A = M_X.$$

The jet map j_h assigns the Taylor class through order $m_\rho - 1$ at every ρ . Here is an explicit justification of its polynomial coordinate interpretation. If $H_\rho(X) = h(X)/(X - \rho)^{m_\rho}$, then a prescribed germ f_ρ in the ρ -block is inserted, with all other blocks zero, by

$$\left[H_\rho(X) \sum_{j=0}^{m_\rho-1} \frac{1}{j!} \frac{d^j}{dX^j} \left(\frac{f_\rho(X)}{H_\rho(X)} \right) \Big|_{X=\rho} (X - \rho)^j \right]_h.$$

At ρ , the Taylor truncation has exactly the prescribed jet. At any other selected point, H_ρ vanishes to the full required order. The formula is unchanged if f_ρ changes by a germ of order at least m_ρ . Summing these insertions realizes every jet packet. A polynomial has zero packet exactly when it is divisible by every $(X - \rho)^{m_\rho}$, hence by their product h . This proves

$$E_h \cong \bigoplus_{\rho \in Z} \mathcal{O}_\rho / ((s - \rho)^{m_\rho}) = \bigoplus_{\rho \in Z} \mathcal{O}_\rho / (g),$$

with the actual coordinates and orders unchanged.

Put

$$v_h = g/h, \quad \varepsilon = j_h(h/g) \in E_h^\times, \quad R_Z(u) = \text{rem}_h(\varepsilon u).$$

Both quotients in these expressions are taken as germs at the selected zeros; complete multiplicity makes them holomorphic units there. In particular $j_h(v_h)\varepsilon = 1$. Define $J_Z = j_h\mathcal{M} : \mathcal{B} \rightarrow E_h$. The factorization gH_ϕ proves $J_Z\Theta = 0$, with all jets, not only values. Thus J_Z descends to $\bar{J}_Z : Q \rightarrow E_h$.

We now prove the inverse used for a section, including its analytic domain. For $F \in \mathcal{B}$ with $\mathcal{M}F(\rho) = 0$, put

$$S_\rho F(x) = x^{-\rho} \int_x^\infty F(y)y^{\rho-1} dy.$$

All complex powers use the real logarithm on $\mathbb{R}_{>0}$. Differentiating gives $(D - \rho)S_\rho F = F$. The zero moment rewrites the upper integral as minus the integral from zero to x . If $N > \Re \rho$ and $N + \Re \rho > 0$, these two formulas give respectively

$$|S_\rho F(x)| \leq \frac{p_{N,0}(F)}{N - \Re \rho} x^{-N}, \quad |S_\rho F(x)| \leq \frac{p_{-N,0}(F)}{N + \Re \rho} x^N.$$

Use the first for $x \geq 1$, the second for $0 < x \leq 1$, and enlarge N for each requested weight. The identity $DS_\rho F = \rho S_\rho F + F$ expresses every Euler derivative of $S_\rho F$ as $\rho^j S_\rho F$ plus a finite linear combination of Euler derivatives of F . This proves $S_\rho F \in \mathcal{B}$ and continuity on the specified closed moment kernel. The homogeneous solution $cx^{-\rho}$ belongs to $\mathcal{B}$ only if $c = 0$, by considering both ends and all integer weights. Thus this inverse is unique. Taking Mellin transforms gives

$$\mathcal{M}S_\rho F(s) = \frac{\mathcal{M}F(s)}{s - \rho},$$

with the removable value filled in.

Iterate once for each prescribed vanishing factor. Division lowers that zero order by one and preserves all other required zeros. This constructs the continuous inverse S_h of $h(D)$ on $\ker J_Z$. Conversely, the Mellin transform of $h(D)F$ is $h(s)\mathcal{M}F(s)$, so lies in that kernel. Each $D - \rho$ is injective on $\mathcal{B}$ by the homogeneous-solution calculation. Hence

$$h(D)\mathcal{B} = \ker J_Z, \quad h(D) : \mathcal{B} \longrightarrow \ker J_Z \text{ is a continuous bijection with inverse } S_h.$$

Define the actual source section

$$R : E_h \longrightarrow \mathcal{B}, \quad Ru = S_h \Theta(R_Z(u)(D)\phi_*).$$

Its input has Mellin transform $gR_Z(u)$, hence belongs to the full required jet kernel. The inverse calculation proves

$$\mathcal{M}Ru = v_h R_Z(u), \quad J_Z Ru = j_h(v_h)\varepsilon u = u.$$

Thus, for $q_Q : \mathcal{B} \rightarrow Q$,

$$\sigma = q_Q R : E_h \hookrightarrow Q, \quad \bar{J}_Z \sigma = 1, \quad h(D)\sigma = 0.$$

The last equality follows because $h(D)Ru = \Theta(R_Z(u)(D)\phi_*)$, an original theta boundary. The left inverse proves injectivity. This section contains the full arithmetic unit ε ; it is not obtained by setting that unit to 1.

A.8 The exact scaling defect before quotienting

Set $\ell(u) = [X^{d_h-1}]R_Z(u)$. The polynomial $XR_Z(u) - R_Z(Au)$ vanishes modulo h , has degree at most d_h , and has leading coefficient $\ell(u)$. Since h is monic, this proves

$$XR_Z(u) - R_Z(Au) = h(X)\ell(u).$$

The operators D and Θ commute by termwise differentiation. Apply $h(D)$ to $DR - RA$, use the preceding polynomial identity, and compare with $h(D)f_0\ell$. Injectivity of $h(D)$ proves the full source identity

$$DR - RA = f_0\ell.$$

For one simple zero this coefficient is $\ell(u) = u/g'(\rho) = u/(2\xi'(\rho))$.

For $a > 0$, let $U_a F(x) = F(x/a)$. It preserves both original function spaces and satisfies $U_a \Theta = \Theta U_a$. Substitution in the Fourier integral gives $\mathcal{F}U_a = aU_{1/a}\mathcal{F}$. Consequently the original chain action is

$$\mathcal{U}_a^0(\phi, \psi) = (U_a \phi, aU_{1/a} \psi), \quad \mathcal{U}_a^1 F = U_a F,$$

and $d\mathcal{U}_a^0 = \mathcal{U}_a^1 d$.

For $t \in \mathbb{R}$, put $T_t = e^{tA}$ and

$$c_t(u) = \int_0^t U_{e^{t-r}} \phi_* \ell(T_r u) dr.$$

The integral is oriented, so it is negative the integral from t to zero when $t < 0$. For the Schwartz seminorm $q_{N,j}(\phi) = \sup_x (1 + |x|)^N |\partial_x^j \phi(x)|$, direct substitution gives

$$q_{N,j}(U_b \phi) \leq b^{-j} \max(1, b)^N q_{N,j}(\phi).$$

Using any fixed norm on the finite-dimensional E_h , only for this estimate, yields

$$q_{N,j}(c_t u) \leq |t| e^{(N+j+\|A\|)|t|} q_{N,j}(\phi_*) \|\ell\| \|u\|.$$

The integrand is continuous in each seminorm. Its Riemann sums therefore converge in the complete closed Schwartz subspace V , proving that the integral is an actual vector of that source. The analogous identity $p_{b,j}(U_a F) = a^b p_{b,j}(F)$ controls the target space. Differentiation of dilations in each of these seminorms follows from the integral remainder for the derivative and the same bounds applied to $D^2 \phi$ or $D^2 F$. Thus, in $\mathcal{B}$,

$$\frac{d}{dr} (U_{e^{t-r}} R T_r u) = -U_{e^{t-r}} (DR - RA) T_r u = -\Theta U_{e^{t-r}} \phi_* \ell(T_r u).$$

Integration proves, including the sign for negative t ,

$$U_{e^t} R - R T_t = \Theta c_t.$$

Inserting $U_{e^t} R T_s$ into the defect for $t + s$, and then using injectivity of Θ , proves the cocycle identity

$$c_{t+s} = U_{e^t} c_s + c_t T_s, \quad c_0 = 0.$$

In the two-leg complex the two literal homotopies are

$$H_t^+(u) = (c_t(u), 0), \quad H_t^-(u) = (0, -\widehat{c_t(u)}).$$

Both have differential Θc_t . Their difference $(c_t, \widehat{c_t})$ is the degree-zero cycle obtained from the Fourier graph; it has not been discarded. After quotienting,

$$\overline{U}_a \sigma = \sigma a^A \quad (a > 0).$$

This proves equivariance of the section in Q and states exactly the retained boundary at the source level.

A.9 The cited finite-field theorem and the full-jet comparison

For clarity, the classical theorem being used as an inspiration has specific geometric hypotheses. Let $\mathbb{F}_q$ be a finite field of characteristic p , let X_0 be smooth and projective over it, let $X = X_0 \times_{\mathbb{F}_q} \overline{\mathbb{F}_q}$, and let $\ell \neq p$ be prime. Deligne's theorem states that the Frobenius

characteristic polynomial on $H_{\text{ét}}^i(X, \mathbb{Q}_\ell)$ has integral coefficients independent of ℓ , and that every complex conjugate of each eigenvalue has absolute value $q^{i/2}$ (Deligne, 1974, theorem (1.6), p. 276). Here the Frobenius action is the one in Deligne's statement: the pullback induced by the q -power geometric endomorphism, equivalently the geometric Galois Frobenius with his convention. The finite-field cardinality, the geometric cohomology, and this action are part of the theorem. Deligne's later theory of weights is developed in Deligne (1980). These citations acknowledge his results; they do not assert that the present chart and Schwartz source already satisfy those geometric hypotheses.

Exposé-level credits behind this literature are retained in the source guide. In particular, SGA 7, Exposé XXI is by Katz, with its integrality appendix by Deligne (Katz, 1973; Deligne, 1973). Illusie's account records the rédaction and contributor roles in SGA 5 and SGA 4 $\frac{1}{2}$ (Illusie, n.d.; Deligne, 1977). These historical credits do not add hypotheses to the theorem above.

The actual local comparison in this programme is as follows. On a selected zero retain its coordinate $z = s - \rho$ and the complete germ

$$g_\rho(z) = g(\rho + z) = u_\rho(z)z^{m_\rho}, \quad u_\rho(0) \neq 0, \quad A_\rho = \mathcal{O}_\rho/(g_\rho).$$

The equality of ideals $(g_\rho) = (z^{m_\rho})$ follows from invertibility of u_ρ ; it supplies Taylor coordinates but does not remove the unit from any formula. For a real parameter $q > 0$, define

$$C_q f(z) = f(z/q) \pmod{g_\rho}, \quad N_\rho f = zf.$$

This q is a positive grading parameter, not a claim of finite-field structure. In particular it must not be identified solely by its notation with the polynomial degree denoted q elsewhere. The substitution is well-defined because

$$\frac{g_\rho(z/q)}{g_\rho(z)} = q^{-m_\rho} \frac{u_\rho(z/q)}{u_\rho(z)}$$

is a holomorphic unit at zero. Its inverse is substitution $z \mapsto qz$. Thus $C_p C_q = C_{pq}$, $C_1 = 1$, and $C_q(z^j) = q^{-j} z^j$ for $0 \leq j < m_\rho$.

The original Mellin scaling acts on the same full block by

$$U_a|_{A_\rho} = a^\rho e^{(\log a)N_\rho} = a^\rho \sum_{j=0}^{m_\rho-1} \frac{(\log a)^j}{j!} N_\rho^j, \quad a > 0,$$

because $\mathcal{M}(U_a F)(s) = a^s \mathcal{M}F(s)$. Direct substitution and multiplication now give

$$C_q N_\rho C_q^{-1} = q^{-1} N_\rho, \quad U_a N_\rho U_a^{-1} = N_\rho,$$

$$C_q U_a|_{A_\rho} C_q^{-1} = a^\rho e^{(\log a)N_\rho/q} = a^{\rho(1-1/q)} U_{a^{1/q}}|_{A_\rho}.$$

These are exact morphism identities on the original local algebra, retaining the complete finite exponential and its scalar factor.

There is also a precise obstruction to a particular simultaneous identification. Write $A_Z = \bigoplus_{\rho \in Z} A_\rho$, $N = \bigoplus N_\rho$, and retain U_a on that sum. Suppose a target vector space carries an invertible F and an operator N_D satisfying $F N_D F^{-1} = q^{-1} N_D$, with $q \neq 1$. For every linear map T intertwining both proposed operators, namely $T U_a = F T$ and $T N = N_D T$, we have $F^{-1} T = T U_a^{-1}$, and therefore

$$q^{-1} N_D T = F N_D F^{-1} T = F T N U_a^{-1} = T U_a N U_a^{-1} = T N = N_D T.$$

It follows that $TN = 0$. The unique surviving factorization is through

$$A_Z \twoheadrightarrow A_Z/NA_Z \cong \bigoplus_{\rho \in Z} \mathbb{C}, \quad [f_\rho] \mapsto (f_\rho(0))_{\rho \in Z}.$$

The kernel of evaluation in each Taylor block is exactly $zA_\rho = N_\rho A_\rho$, proving the quotient identification. Evaluation intertwines U_a with $\text{diag}(a^\rho)$ and sends N to zero, giving a concrete surviving comparison. An injective simultaneous intertwiner can exist only when all $N_\rho = 0$, equivalently all retained orders are one.

This obstruction concerns those two specified intertwining equations. The full-jet automorphism C_q remains available and has already been constructed above. Its conjugacy relation alone cannot supply Deligne’s absolute weight conclusion: for any nonzero scalar c , replacing C_q by cC_q leaves the conjugacy with N unchanged and multiplies every eigenvalue by c . On a simple block $C_q = 1$ for every q , whereas the original scaling still has eigenvalue a^ρ . The displayed source sections, support maps, and operator comparisons are the proved connections. Further geometric and quantitative constructions remain part of the research programme, with their original objects and hypotheses still present.

B Reading the human-source record

The source record follows a mathematical use, rather than ending at the most recent project note that contains it. It records the cited work, its human authors and other documented contributors, the precise result used, the programme’s receiving statement, and the evidence actually inspected. An earlier source credited by that work receives a further dependency edge when its contribution is relevant to the argument. Authors of an exposition and authors of its underlying results retain their respective credits.

The accompanying files distinguish five levels of evidence. A *primary-text check* means the stated passage of the human source was read and compared with its application. An *authenticated identity* establishes the publication and author metadata, without implying that its theorem was read. An *inherited review* preserves an older source record with its original scope. A *discovery candidate* is a name, eponym, formula, or citation awaiting resolution. An *unresolved edge* states the missing evidence. These labels are attached to specific entries; they are not interchangeable quality scores for an entire paper.

This distinction matters for the present corpus. A scan of the 692-page predecessor’s 15 editable text files covered 223,268 lines. Its bibliography blocks contain five distinct programme references and no human-authored bibliography entry. This does not show that human sources were never used: the historical literature catalogue and recovered reading records contain many such sources. The companion reconnects those sources to the actual mathematical uses. A name search by itself establishes neither a dependency nor a theorem’s correctness.

The original acquired public conversation has 5,349 nonempty message records and 5,350 nodes including its empty root. Its 215 readable conversational records are a selection; 2,600 tool payloads were redacted in the published chain. Those missing bodies have not been reconstructed merely because a reference survives. Other exports and retained mathematical manuscripts are audited separately, with duplicate bodies identified where possible. `COVERAGE.md` records the current boundary and remaining source gaps. `ATTRIBUTE_MAP.md` and `ATTRIBUTE_GRAPH.json` connect uses to source works; `UNRESOLVED_ATTRIBUTE.md` gives the exact remaining checks. The six tau reports and the expanded historical bibliography preserve the longer recursive source trails.

B.1 A worked attribution repair: the logarithmic cumulant identity

The secondary globalization note uses an unnamed standard moment–cumulant identity. Its exact formula is [McCullagh \(2018, §2.3.4, p. 38, equation \(2.9\)\)](#), whose formal-series derivation applies without positivity assumptions. A further human reading route is [McCullagh \(2011, p. 2\)](#), which explains the partition-lattice treatment of [Speed \(1983\)](#) and its connection with the inversion framework of [Rota \(1964\)](#). McCullagh’s text was read; the two earlier publication identities were authenticated separately. Their complete original articles were not newly read in this check. The following calculation supplies the exact convention map to the project formula ([The Clankers, 2026](#)).

Let F be analytic and nonzero near t , and write

$$m_n = \frac{F^{(n)}(t)}{F(t)}, \quad \kappa_n = (\log F)^{(n)}(t), \quad M(z) = \frac{F(t+z)}{F(t)} = 1 + \sum_{j \geq 1} m_j \frac{z^j}{j!}.$$

A local analytic logarithm exists. Its positive-order derivatives are independent of the chosen branch. Expanding its Taylor series gives

$$\sum_{n \geq 1} \kappa_n \frac{z^n}{n!} = \log M(z) = \sum_{r \geq 1} \frac{(-1)^{r-1}}{r} \left(\sum_{j \geq 1} m_j \frac{z^j}{j!} \right)^r.$$

For ordered positive sizes $j_1 + \dots + j_r = n$, the coefficient $n!/(j_1! \dots j_r!)$ counts ordered partitions of $\{1, \dots, n\}$ into blocks of those sizes. Each unordered partition with r blocks occurs $r!$ times. The factor $1/r$ therefore leaves $(r-1)!$, proving

$$\kappa_n = \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi|-1)! \prod_{B \in \pi} m_{|B|}.$$

In the note’s notation, $A_j(t; s) = (s)_j L_t(s+j)/L_t(s)$ and $m_j = (-1)^j A_j$. For each partition, the block sizes sum to n . Consequently its full sign is $(-1)^n$, and the exact receiving identity is

$$\partial_t^n \log L_t(s) = (-1)^n \sum_{\pi \in \Pi_n} (-1)^{|\pi|-1} (|\pi|-1)! \prod_{B \in \pi} A_{|B|}(t; s).$$

This is a classical combinatorial identity applied in the note’s analytic domain. The complex-valued application requires no probability measure. The source attribution has been repaired; no priority claim follows from the project specialization.

B.2 How to use the companion files

`HUMAN_SOURCES.md` provides the bibliographic and mathematical role of the resolved sources; `REFERENCES.bib` retains their reusable identities. The attribution records include project-source edges and source-to-source edges with verification levels. The inherited bibliography is retained as provenance without upgrading all of it to a fresh primary reading. Historical routes no longer used in the current calculation remain credited with their historical role.

The public guide excludes private conversation and assignment text. Source hashes and stable public theorem locations allow a mathematical comparison without publishing those private records. A source gap remains visible until its missing identity, passage, or application map is actually resolved. The mathematical tasks continue while that reconstruction proceeds.

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